

A Course Based Project on
Analog Frequency Based Security System
Submitted to the Department of ECE
in partial fulfillment of the requirements for the completion of course
ANALOG CIRCUITS LABORATORY (22PC2EC205)
BACHELOR OF TECHNOLOGY
in
ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

N. Siddhartha	24071A04H4
P.Sai Manaswi	24071A04H5
P.Swetha	24071A04H6
P.Nithin Reddy	25075A04H7

UNDER THE SUPERVISION OF

Mrs.Sri Rama, Assistant Professor
Dr.K.Aruna Kumari, Assistant Professor



VNRVJiet

VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

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VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI
INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Electronics and Communication Engineering



CERTIFICATE

This is to certify that the course based project entitled “**Analog Frequency Based Security System**” is being submitted by **N.Siddhartha(24071A04H4)**, **P.Sai_Manaswi(24071A04H5)**, **P.Swetha (25075A04H6)**, **P.Nithin_Reddy(24071A04H7)**, for the partial fulfillment of requirements for course based project of **Analog Circuits Laboratory** in II year II semester of **Bachelor of Technology** in Electronics and Communication Engineering of VNRVJIET, Hyderabad during the academic year 2025 - 2026.

Course Coordinator

Dr.K.Aruna Kumari

Assistant Professor
VNRVJIET, Hyd.

Course Coordinator

Mrs. Sri Rama

Assistant Professor
VNRVJIET, Hyd.

Head of the Department

Dr. P. Kishore

Assoc. Professor & Head
VNRVJIET, Hyd.

DECLARATION

We do declare that the course based project report entitled “**Analog Frequency Based Security System**” submitted to the Department of Electronics and Communication Engineering, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirements for course based project of **Analog Circuits Laboratory** in II B.Tech. II Semester for the academic year 2025 - 2026.

Place: Hyderabad

Date:

	Student Name	Roll No.	Student Signature
1.	N. Siddhartha	24071A04H4	
2.	P.Sai Manaswi	24071A04H5	
3.	P.Swetha	24071A04H6	
4.	P.Nithin Reddy	25075A04H7	

Verified by:

Dr. K. Aruna Kumari
Assistant Professor

Mrs. Sri Rama
Assistant Professor

Date of Verification:

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ABSTRACT

This project presents an analog frequency-based security system that uses a specific sound frequency as a key to unlock a system. Unlike traditional security systems that rely on passwords or digital authentication, this design uses an audio signal of a particular frequency (such as 1 kHz) for access control. Only when the correct frequency is detected, the system activates an output device like a relay or LED.

The system begins with an electret microphone that captures the input sound and converts it into an electrical signal. This signal is usually weak, so it is amplified using an operational amplifier. The amplified signal is then passed through a band-pass filter, which allows only the desired frequency to pass while blocking unwanted signals and noise.

A comparator circuit is used to compare the filtered signal with a reference level. If the detected frequency matches the required value, the comparator produces a high output signal. This signal is then given to a 555 timer configured in monostable mode, which generates a stable output pulse for a fixed duration.

Finally, the output from the 555 timer drives a relay or an indicator such as an LED or buzzer. This completes the unlocking mechanism of the system. The project demonstrates a low-cost, reliable, and noise-resistant analog security solution without the use of microcontrollers.

1. INTRODUCTION

Security systems are an essential part of modern life, used in homes, offices, banks, and industries to protect valuable assets. Most commonly used systems today are based on digital technologies such as passwords, biometric authentication, and smart cards. While these systems are effective, they can be complex and expensive for small-scale applications.

Analog systems, on the other hand, offer a simpler and cost-effective alternative for basic security needs. This project introduces a unique analog approach where a specific sound frequency acts as a password. The concept is based on the principle that a particular frequency can be identified and used to trigger a response, making it suitable for secure access control.

The system consists of several stages including sound detection, signal amplification, filtering, comparison, and output activation. Each stage plays an important role in ensuring that only the correct frequency is accepted. The band-pass filter is especially important as it eliminates unwanted frequencies and improves the accuracy of detection.

This project not only provides a practical security solution but also helps in understanding fundamental concepts of analog electronics such as filters, amplifiers, and timers. It can be applied in areas like door locking systems, lockers, and anti-theft devices, making it a useful and educational project for students.

2. BLOCK DIAGRAM

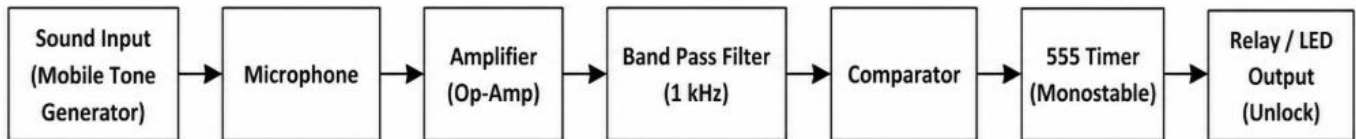
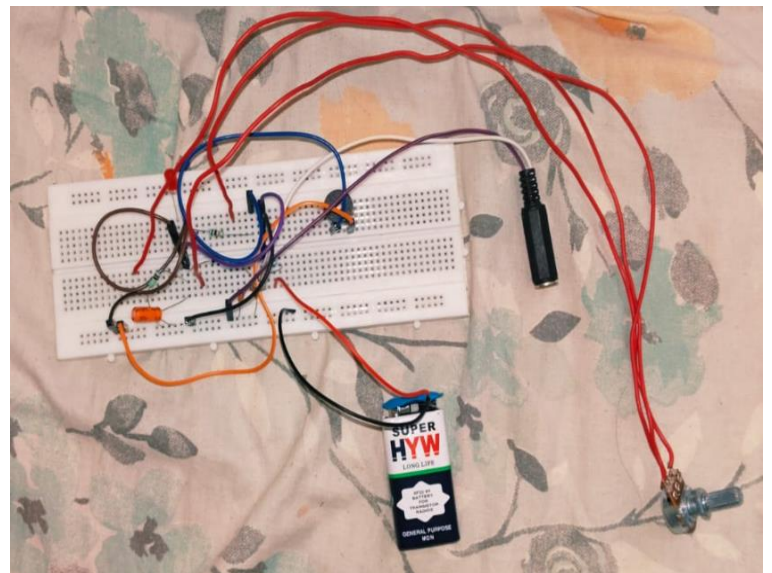
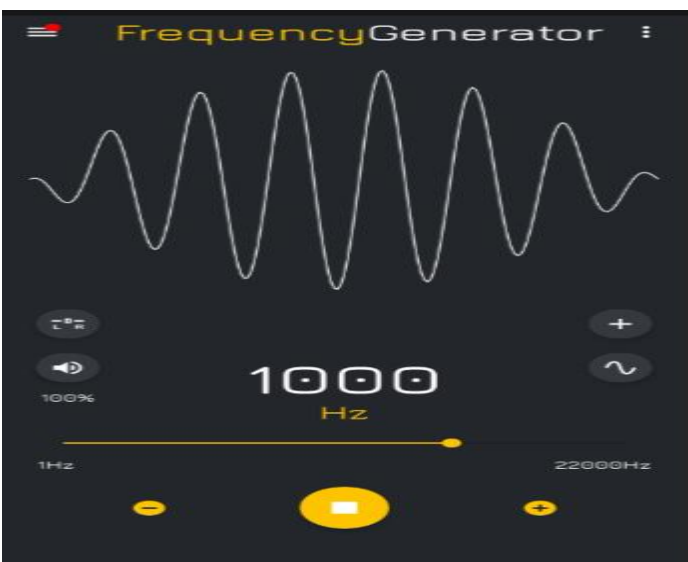


Fig 2.1: Analog frequency based security system

3. WORKING

The working of the analog frequency-based security system begins with the electret microphone, which captures the input sound signal from a tone generator or mobile phone. This sound signal is converted into a weak electrical signal, which is then amplified using an operational amplifier to make it suitable for further processing. The amplified signal is passed through a band-pass filter designed to allow only a specific frequency (around 1 kHz) while rejecting all other unwanted frequencies and noise.

The filtered signal is then given to a comparator circuit, where it is compared with a preset reference voltage. If the input signal matches the desired frequency, the comparator produces a HIGH output. This output triggers the 555 timer configured in monostable mode, which generates a fixed-duration pulse. The pulse is then used to drive a relay or indicator such as an LED or buzzer, completing the unlocking mechanism.



4.OUTPUT

The output of the system depends on the input frequency applied. When the correct frequency (e.g., 1 kHz) is given through the sound source, the system successfully detects it and activates the output. As a result, the relay switches ON or the LED glows, indicating that access is granted or the system is unlocked.

If any other frequency or random noise is applied, the band-pass filter blocks it, and the comparator does not produce a triggering signal. Therefore, the output remains OFF, ensuring that only the correct frequency can activate the system. This makes the system selective and secure.

5.CONCLUSION

The analog frequency-based security system successfully demonstrates a unique method of access control using sound signals. By using a specific frequency as a key, the system ensures that only the correct input can activate the output, making it a reliable and selective security mechanism. This approach provides an alternative to conventional password or digital systems.

The project effectively utilizes basic analog components such as microphones, operational amplifiers, filters, comparators, and timers. Each stage plays an important role in processing the signal, from detecting sound to filtering and finally triggering the output. This shows how fundamental analog electronics concepts can be applied in practical applications.

One of the key advantages of this system is its simplicity and low cost. It does not require complex programming or microcontrollers, making it easy to design, build, and understand. Additionally, the use of a band-pass filter helps in rejecting unwanted noise, improving the accuracy and reliability of the system.

Overall, this project provides a strong foundation in analog circuit design and demonstrates a practical application in the field of security systems. It can be further enhanced by adding multiple frequency detection or improving sensitivity, making it a flexible and extendable solution for real-world applications.

6. REFERENCES

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